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Database Development for a Mobile application to Collect Dysmenorrhea Data

1. System intent

This project is centered on the development of a database for a mobile application focused on collecting and managing the data for the pathology dysmenorrhea. The system is aimed to register and store the user's symptoms, menstrual cycle information and the biomarkers, for a future data analysis or integration of artificial intelligence. The database for dysmenorrhea is built from a previous Android application, in order to extend it by utilizing a structured and scalable database. The work consists in transforming the application requirements into a conceptual data model, Entity-Relationship (ER) diagram, and converting this into a relational schema which is implemented by using SQL

2. Project features

1. User registration and identification

The system allows each user to be identified and associated with their records.

2. Storage of personal and clinical background information

The system stores demographic and gynecological information such as age, weight, height, infertility, family history, age at menarche, and nulliparity.

3. Record menstrual cycles

The system allows each user to register multiple menstrual cycles.

4. Track daily symptoms

The system stores daily records within each cycle, including pain intensity, menstrual flow, and date.

5. Lifestyle tracking

The system stores daily factors such as stress, depression, diet, alcohol consumption, exercise, smoking, oral contraceptives.

6. Record treatments

The system stores medication and non-pharmacological treatments.

7. View and update user data

The system allows users to view, load, and update their personal, cycle, and daily records.

8. View cycle history and daily records.

The system allows users to view their cycle history and associated daily records.

9. Manage multiple cycles per user.

The system must support the management of multiple cycles per user.

10. Enable data extraction for analysis.

The system must allow stored data to be extracted for analysis and future processing.

3. Requirements table

ID	Requirement Type	Requirement	Priority
FR1	Functional	The system shall store user personal information for each user.	High
FR2	Functional	The system shall store gynecological background data for each user.	High
FR3	Functional	The system shall allow one user to register multiple menstrual cycles.	High
FR4	Functional	The system shall allow each cycle to contain multiple daily entries.	High
FR5	Functional	The system shall store daily dysmenorrhea-related symptoms.	High
FR6	Functional	The system shall store daily lifestyle-related variables.	High
FR7	Functional	The system shall allow updating previously stored data.	High
FR8	Functional	The system shall retrieve stored user and cycle data when needed.	High
FR9	Functional	The system shall maintain relationships between users, cycles, and daily entries.	High
NFR1	Non-functional	The database shall ensure data integrity through primary and foreign keys.	High
NFR2	Non-functional	The system shall support scalability for storing data from multiple users.	Medium
NFR3	Non-functional	The system shall use a structured format suitable for future analytics.	High
NFR4	Non-functional	The system should reduce dependence on local file-based storage.	High

4. Main Actors

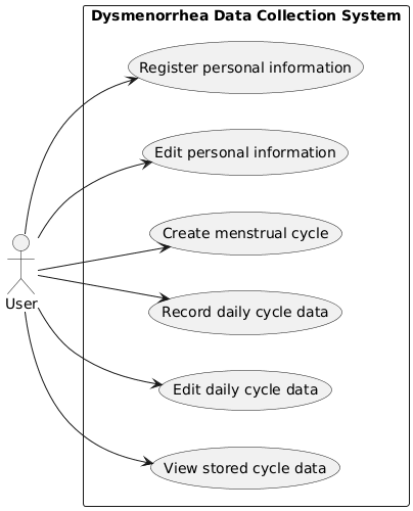
The only actor is going to be the User (patient).

```
USER
 └── CYCLES
     └── DAYS
         └── symptoms
```

5. Use Cases

Use Case ID	Name
UC1	Register Personal Information
UC2	Edit Personal Information
UC3	Create Menstrual Cycle
UC4	Record Daily Symptoms
UC5	Record Lifestyle Factors
UC6	Record Treatments
UC7	Edit Daily Cycle Data
UC8	View Cycle History
UC9	View Stored Data

5.1 Use Case Diagram

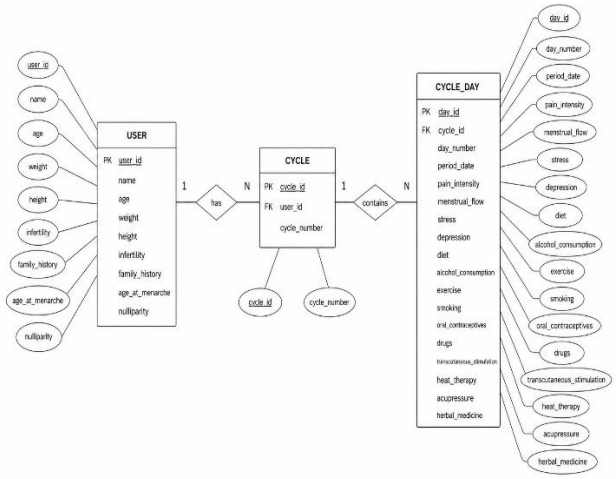


6. **Traceability Matrix**

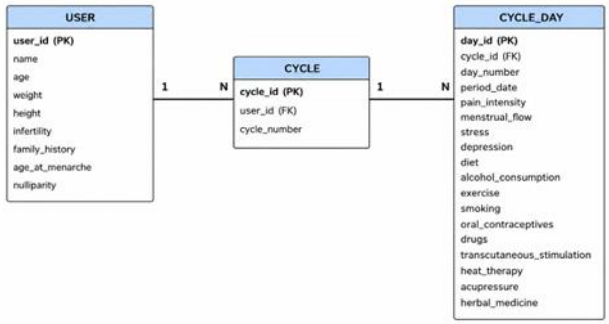
Requirement ID	**Description**	**UC1 Register Personal Information**	**UC2 Edit Personal Information**	**UC3 Create Menstrual Cycle**	**UC4 Record Daily Cycle Data**	**UC5 Edit Daily Cycle Data**	**UC6 View Stored Data**
FR1	Store personal information for each user	X	X				X
FR2	Store gynecological background data for each user	X	X				X
FR3	Allow each user to register multiple menstrual cycles			X			X
FR4	Allow each cycle to contain multiple daily entries			X	X	X	X
FR5	Store daily dysmenorrhea-related symptoms				X	X	X
FR6	Store daily lifestyle-related variables				X	X	X
FR7	Allow previously stored data to be updated		X			X	
FR8	Retrieve stored user and cycle data						X
FR9	Maintain relationships between users, cycles, and daily entries	X	X	X	X	X	X
NFR1	Ensure data integrity through primary and foreign keys	X	X	X	X	X	X
NFR2	Support scalability for storing data from multiple users	X	X	X	X	X	X
NFR3	Provide a structured format suitable for future analysis	X	X	X	X	X	X

7. ER Diagram

To transform this structure into a relational database, the information was normalized into three entities: USER, CYCLE, and CYCLE_DAY. A user can have multiple cycles, and each cycle can include multiple daily entries. This design preserves the application logic while improving scalability, consistency, and suitability for future data analysis and AI applications.



8. Relational Model



9. SQL DDL

```
CREATE TABLE USER (
    user_id INTEGER PRIMARY KEY AUTOINCREMENT,
    name VARCHAR(100),
    age INTEGER,
    weight DECIMAL(5,2),
    height DECIMAL(5,2),
    infertility BOOLEAN,
    family_history BOOLEAN,
```

```
    age_at_menarche INTEGER,
    nulliparity BOOLEAN
);

CREATE TABLE CYCLE (
    cycle_id INTEGER PRIMARY KEY AUTOINCREMENT,
    user_id INTEGER NOT NULL,
    cycle_number INTEGER,
    FOREIGN KEY (user_id) REFERENCES USER(user_id)
);

CREATE TABLE CYCLE_DAY (
    day_id INTEGER PRIMARY KEY AUTOINCREMENT,
    cycle_id INTEGER NOT NULL,
    day_number INTEGER,
    period_date DATE,
    pain_intensity INTEGER,
    menstrual_flow VARCHAR(20),
    stress INTEGER,
    depression INTEGER,
    diet VARCHAR(100),
    alcohol_consumption VARCHAR(50),
    exercise VARCHAR(50),
    smoking BOOLEAN,
    oral_contraceptives BOOLEAN,
    drugs VARCHAR(100),
    transcutaneous_stimulation BOOLEAN,
    heat_therapy BOOLEAN,
    acupressure BOOLEAN,
    acupuncture BOOLEAN,
    herbal_medicine VARCHAR(100),
    FOREIGN KEY (cycle_id) REFERENCES CYCLE(cycle_id)
);
```

10. DML

To validate the database design, sample records were inserted into the USERS, CYCLE, and CYCLE_DAY tables. Several SQL queries were executed to verify that the data was stored correctly and that the relationships between tables worked as expected.

```
INSERT INTO USER (
    user_id, name, age, weight, height, infertility, family_history, age_at_menarche, nulliparity
) VALUES (
    1, 'Ana Lopez', 21, 58.5, 165.0, 0, 1, 12, 1
);
```

```
INSERT INTO CYCLE (
    cycle_id, user_id, cycle_number
) VALUES (
    1, 1, 1
);

INSERT INTO CYCLE_DAY (
    day_id, cycle_id, day_number, period_date, pain_intensity, menstrual_flow,
    stress, depression, diet, alcohol_consumption, exercise, smoking,
    oral_contraceptives, drugs, transcutaneous_stimulation, heat_therapy,
    acupressure, herbal_medicine
) VALUES (
    1, 1, 1, '2026-04-01', 8, 'Heavy',
    7, 4, 'Balanced', 'No', 'Light', 'No',
    0, 'Ibuprofen', 0, 1,
    0, 'Chamomile'
);

INSERT INTO CYCLE_DAY (
    day_id, cycle_id, day_number, period_date, pain_intensity, menstrual_flow,
    stress, depression, diet, alcohol_consumption, exercise, smoking,
    oral_contraceptives, drugs, transcutaneous_stimulation, heat_therapy,
    acupressure, herbal_medicine
) VALUES (
    2, 1, 2, '2026-04-02', 6, 'Medium',
    5, 3, 'Balanced', 'No', 'Moderate', 'No',
    0, 'Ibuprofen', 1, 1,
    1, 'Chamomile'
);
INSERT INTO USER VALUES (2, 'Marianna Benavides', 24, 60.0, 167.0, 0, 1, 13, 0);

INSERT INTO USER VALUES (3, 'Daniela Romoleroux', 22, 57.0, 162.0, 0, 0, 12, 1);

INSERT INTO USER VALUES (4, 'Carolina Santos', 26, 65.0, 170.0, 1, 1, 14, 0);

INSERT INTO CYCLE VALUES (2, 2, 1);  -- Marianna
INSERT INTO CYCLE VALUES (3, 3, 1);  -- Daniela
INSERT INTO CYCLE VALUES (4, 4, 1);  -- Carolina

-- Cycle_Day for Marianna
INSERT INTO CYCLE_DAY VALUES
(3, 2, 1, '2026-04-05', 7, 'Heavy', 6, 4, 'Balanced', 'No', 'Light', 'No', 0, 'Ibuprofen', 0, 1, 0,
'Chamomile');

INSERT INTO CYCLE_DAY VALUES
```

(4, 2, 2, '2026-04-06', 5, 'Medium', 4, 3, 'Healthy', 'Yes', 'None', 'No', 0, 'None', 1, 1, 0, 'None');

-- Cycle_Day for Daniela
INSERT INTO CYCLE_DAY VALUES
(5, 3, 1, '2026-04-10', 9, 'Heavy', 8, 6, 'Poor', 'Yes', 'None', 'Yes', 1, 'Paracetamol', 0, 1, 1, 'Mint');

INSERT INTO CYCLE_DAY VALUES
(6, 3, 2, '2026-04-11', 7, 'Medium', 6, 5, 'Balanced', 'No', 'Moderate', 'No', 1, 'Paracetamol', 1, 1, 0, 'Chamomile');

-- Cycle_Day for Carolina
INSERT INTO CYCLE_DAY VALUES
(7, 4, 1, '2026-04-15', 6, 'Light', 5, 2, 'Balanced', 'No', 'Moderate', 'No', 0, 'None', 0, 1, 1, 'None');

INSERT INTO CYCLE_DAY VALUES
(8, 4, 2, '2026-04-16', 4, 'Light', 3, 2, 'Healthy', 'No', 'Moderate', 'No', 0, 'None', 0, 0, 0, 'None');

user_id	name	age	weight	height	infertility	family_history	age_at_menarche	nulliparity
1	Ana Lopez	21	58.5	165	0	1	12	1
2	Marianna Benavides	24	60	167	0	1	13	0
3	Daniela Romoleroux	22	57	162	0	0	12	1
4	Carolina Santos	26	65	170	1	1	14	0

cycle_id	user_id	cycle_number
1	1	1
2	2	1
3	3	1
4	4	1

day_id	cycle_id	day_number	period_date	pain_intensity	menstrual_flow	stress	depression	diet	alcohol_consumption	exercise	smoking	oral_contraceptives	drugs	transcutaneous_stimulation	heat_therapy	acupressure	acupuncture	herbal_medicine
1	1	1		8	Heavy	7	4	Balanced	No	Light		0	Ibuprofen	0	1	0		Chamomile
2	1	2		6	Medium	5	3	Balanced	No	Moderate		0	Ibuprofen	1	1	1		Chamomile

INSERT INTO CYCLE_DAY (
day_id, cycle_id, day_number, period_date, pain_intensity, menstrual_flow,
stress, depression, diet, alcohol_consumption, exercise, smoking,
oral_contraceptives, drugs, transcutaneous_stimulation, heat_therapy,

```
    acupressure, herbal_medicine
) VALUES (
    3, 2, 1, '2026-04-05', 7, 'Heavy',
    6, 4, 'Balanced', 'No', 'Light', 'No',
    0, 'Ibuprofen', 0, 1,
    0, 'Chamomile'
);

INSERT INTO CYCLE_DAY (
    day_id, cycle_id, day_number, period_date, pain_intensity, menstrual_flow,
    stress, depression, diet, alcohol_consumption, exercise, smoking,
    oral_contraceptives, drugs, transcutaneous_stimulation, heat_therapy,
    acupressure, herbal_medicine
) VALUES (
    4, 2, 2, '2026-04-06', 5, 'Medium',
    4, 3, 'Healthy', 'Yes', 'None', 'No',
    0, 'None', 1, 1,
    0, 'None'
);

INSERT INTO CYCLE_DAY (
    day_id, cycle_id, day_number, period_date, pain_intensity, menstrual_flow,
    stress, depression, diet, alcohol_consumption, exercise, smoking,
    oral_contraceptives, drugs, transcutaneous_stimulation, heat_therapy,
    acupressure, herbal_medicine
) VALUES (
    5, 3, 1, '2026-04-10', 9, 'Heavy',
    8, 6, 'Poor', 'Yes', 'None', 'Yes',
    1, 'Paracetamol', 0, 1,
    1, 'Mint'
);

INSERT INTO CYCLE_DAY (
    day_id, cycle_id, day_number, period_date, pain_intensity, menstrual_flow,
    stress, depression, diet, alcohol_consumption, exercise, smoking,
    oral_contraceptives, drugs, transcutaneous_stimulation, heat_therapy,
    acupressure, herbal_medicine
) VALUES (
    6, 3, 2, '2026-04-11', 7, 'Medium',
    6, 5, 'Balanced', 'No', 'Moderate', 'No',
    1, 'Paracetamol', 1, 1,
    0, 'Chamomile'
);

INSERT INTO CYCLE_DAY (
    day_id, cycle_id, day_number, period_date, pain_intensity, menstrual_flow,
```

```
    stress, depression, diet, alcohol_consumption, exercise, smoking,
    oral_contraceptives, drugs, transcutaneous_stimulation, heat_therapy,
    acupressure, herbal_medicine
) VALUES (
    7, 4, 1, '2026-04-15', 6, 'Light',
    5, 2, 'Balanced', 'No', 'Moderate', 'No',
    0, 'None', 0, 1,
    1, 'None'
);

INSERT INTO CYCLE_DAY (
    day_id, cycle_id, day_number, period_date, pain_intensity, menstrual_flow,
    stress, depression, diet, alcohol_consumption, exercise, smoking,
    oral_contraceptives, drugs, transcutaneous_stimulation, heat_therapy,
    acupressure, herbal_medicine
) VALUES (
    8, 4, 2, '2026-04-16', 4, 'Light',
    3, 2, 'Healthy', 'No', 'Moderate', 'No',
    0, 'None', 0, 0,
    0, 'None'
);
```

day_id	cycle_id	day_number	period_date	pain_intensity	menstrual_flow	stress	depression	diet	alcohol_consumption	exercise	smoking	oral_contraceptives	drugs	transcutaneous_stimulation	heat_therapy	acupressure	acupuncture	herbal_medicine
1	1	1		8	Heavy	7	4	Balanced	No	Light		0	Ibuprofen	0	1	0		Chamomile
2	1	2		6	Medium	5	3	Balanced	No	Moderate		0	Ibuprofen	1	1	1		Chamomile
3	2	1		7	Heavy	6	4	Balanced	No	Light		0	Ibuprofen	0	1	0		Chamomile
4	2	2		5	Medium	4	3	Healthy	Yes	None		0	None	1	1	0		None
5	3	1		9	Heavy	8	6	Poor	Yes	None		1	Paracetamol	0	1	1		Mint
6	3	2		7	Medium	6	5	Balanced	No	Moderate		1	Paracetamol	1	1	0		Chamomile
7	4	1		6	Light	5	2	Balanced	No	Moderate		0	None	0	1	1		None
8	4	2		4	Light	3	2	Healthy	No	Moderate		0	None	0	0	0		None

- **SELECT QUERIES: used to test if the database is well created.**

Visualize all the users:

```sql
SELECT * FROM USER;
```

	user_id	name	age	weight	height	infertility	family_history	age_at_menarche	nulliparity
1	1	Ana Lopez	21	58.5	165	0	1	12	1
2	2	Marianna Benavides	24	60	167	0	1	13	0
3	3	Daniela Romoleroux	22	57	162	0	0	12	1
4	4	Carolina Santos	26	65	170	1	1	14	0

Visualize all the cycles:

```sql
SELECT * FROM CYCLE;
```

	cycle_id	user_id	cycle_number
1	1	1	1
2	2	2	1
3	3	3	1
4	4	4	1

Visualize all the days of the cycles:

```sql
SELECT * FROM CYCLE_DAY;
```

	day_id	cycle_id	day_number	period_date	pain_intensity	menstrual_flow	stress	depression	diet	alcohol_consumption	exercise	smoking	oral_contraceptives	drugs	transcutaneous_st
1	1	1	1	2026-04-01	8	Heavy	7	4	Balanced	No	Light	No	0	Ibuprofen	0
2	2	1	2	2026-04-02	6	Medium	5	3	Balanced	No	Moderate	No	0	Ibuprofen	1
3	3	2	1	2026-04-05	7	Heavy	6	4	Balanced	No	Light	No	0	Ibuprofen	0
4	4	2	2	2026-04-06	5	Medium	4	3	Healthy	Yes	None	No	0	None	1
5	5	3	1	2026-04-10	9	Heavy	8	6	Poor	Yes	None	Yes	1	Paracetamol	0
6	6	3	2	2026-04-11	7	Medium	6	5	Balanced	No	Moderate	No	1	Paracetamol	1
7	7	4	1	2026-04-15	6	Light	5	2	Balanced	No	Moderate	No	0	None	0
8	8	4	2	2026-04-16	4	Light	3	2	Healthy	No	Moderate	No	0	None	0

- **SELECT with filters**

Visualize the days with high pain:

```sql
SELECT *
FROM CYCLE_DAY
WHERE pain_intensity >= 7;
```

	day_id	cycle_id	day_number	period_date	pain_intensity	menstrual_flow	stress	depression	diet	alcohol_consumption	exercise	smoking	oral_contraceptives	drugs	transcutaneous_st
1	1	1	1	2026-04-01	8	Heavy	7	4	Balanced	No	Light	No	0	Ibuprofen	0
2	3	2	1	2026-04-05	7	Heavy	6	4	Balanced	No	Light	No	0	Ibuprofen	0
3	5	3	1	2026-04-10	9	Heavy	8	6	Poor	Yes	None	Yes	1	Paracetamol	0
4	6	3	2	2026-04-11	7	Medium	6	5	Balanced	No	Moderate	No	1	Paracetamol	1

Visualize the days only with a heavy menstrual flow

```sql
SELECT *
FROM CYCLE_DAY
WHERE menstrual_flow = 'Heavy';
```

	day_id	cycle_id	day_number	period_date	pain_intensity	menstrual_flow	stress	depression	diet	alcohol_consumption	exercise	smoking	oral_contraceptives	drugs	transcutaneous_stimu
1	1	1	1	2026-04-01	8	Heavy	7	4	Balanced	No	Light	No	0	Ibuprofen	0
2	3	2	1	2026-04-05	7	Heavy	6	4	Balanced	No	Light	No	0	Ibuprofen	0
3	5	3	1	2026-04-10	9	Heavy	8	6	Poor	Yes	None	Yes	1	Paracetamol	0

Visualize the days in which they consumed more alcohol

```sql
SELECT *
FROM CYCLE_DAY
WHERE alcohol_consumption = 'Yes';
```

	day_id	cycle_id	day_number	period_date	pain_intensity	menstrual_flow	stress	depression	diet	alcohol_consumption	exercise	smoking	oral_contraceptives	drugs	transcutaneous_stimula
1	4	2	2	2026-04-06	5	Medium	4	3	Healthy	Yes	None	No	0	None	1
2	5	3	1	2026-04-10	9	Heavy	8	6	Poor	Yes	None	Yes	1	Paracetamol	0

- **SELECT with JOIN**

Visualize the complete information per user

```sql
SELECT
    U.name,
    C.cycle_number,
    D.day_number,
    D.period_date,
    D.pain_intensity,
    D.menstrual_flow,
    D.drugs
FROM USER U
JOIN CYCLE C ON U.user_id = C.user_id
JOIN CYCLE_DAY D ON C.cycle_id = D.cycle_id
ORDER BY U.name, C.cycle_number, D.day_number;
```

	name	cycle_number	day_number	period_date	pain_intensity	menstrual_flow	drugs
1	Ana Lopez	1	1	2026-04-01	8	Heavy	Ibuprofen
2	Ana Lopez	1	2	2026-04-02	6	Medium	Ibuprofen
3	Carolina Santos	1	1	2026-04-15	6	Light	None
4	Carolina Santos	1	2	2026-04-16	4	Light	None
5	Daniela Romoleroux	1	1	2026-04-10	9	Heavy	Paracetamol
6	Daniela Romoleroux	1	2	2026-04-11	7	Medium	Paracetamol
7	Marianna Benavides	1	1	2026-04-05	7	Heavy	Ibuprofen
8	Marianna Benavides	1	2	2026-04-06	5	Medium	None

Visualize an only one user

```sql
SELECT
    U.name,
    C.cycle_number,
    D.day_number,
    D.period_date,
    D.pain_intensity
FROM USER U
JOIN CYCLE C ON U.user_id = C.user_id
JOIN CYCLE_DAY D ON C.cycle_id = D.cycle_id
WHERE U.name = 'Ana Lopez';
```

	name	cycle_number	day_number	period_date	pain_intensity
1	Ana Lopez	1	1	2026-04-01	8
2	Ana Lopez	1	2	2026-04-02	6

Average pain per user

```sql
SELECT
    U.name,
    AVG(D.pain_intensity) AS avg_pain
FROM USER U
JOIN CYCLE C ON U.user_id = C.user_id
JOIN CYCLE_DAY D ON C.cycle_id = D.cycle_id
GROUP BY U.name;
```

	name	avg_pain
1	Ana Lopez	7.0
2	Carolina Santos	5.0
3	Daniela Romoleroux	8.0
4	Marianna Benavides	6.0

Maximum pain registered per user

```sql
SELECT
    U.name,
    MAX(D.pain_intensity) AS max_pain
FROM USER U
JOIN CYCLE C ON U.user_id = C.user_id
JOIN CYCLE_DAY D ON C.cycle_id = D.cycle_id
GROUP BY U.name;
```

	name	max_pain
1	Ana Lopez	8
2	Carolina Santos	6
3	Daniela Romoleroux	9
4	Marianna Benavides	7

Number of days registered per cycle

```sql
SELECT
    cycle_id,
    COUNT(*) AS total_days
FROM CYCLE_DAY
GROUP BY cycle_id;
```

	cycle_id	total_days
1	1	2
2	2	2
3	3	2
4	4	2

- **UPDATE queries**

Change the weight of a user

```sql
UPDATE USER
SET weight = 58.0;


SELECT * FROM USER WHERE user_id = 1;
```

	user_id	name	age	weight	height	infertility	family_history	age_at_menarche	nulliparity
1	1	Ana Lopez	21	58	165	0	1	12	1

Change the pain intensity of a day

```sql
UPDATE CYCLE_DAY
SET pain_intensity = 9
WHERE day_id = 2;
SELECT * FROM CYCLE_DAY WHERE day_id = 2;
```

	day_id	cycle_id	day_number	period_date	pain_intensity	menstrual_flow	stress	depression	diet	alcohol_consumption	exercise	smoking	oral_contraceptives	drugs	transcutaneous_stim
1	2	1	2	2026-04-02	9	Medium	5	3	Balanced	No	Moderate	No	0	Ibuprofen	1

Change the treatment used

```sql
UPDATE CYCLE_DAY
SET drugs = 'Naproxen'
WHERE day_id = 4;
SELECT * FROM CYCLE_DAY WHERE day_id = 4;
```

	day_id	cycle_id	day_number	period_date	pain_intensity	menstrual_flow	stress	depression	diet	alcohol_consumption	exercise	smoking	oral_contraceptives	drugs	transcutaneous_stimulatior
1	4	2	2	2026-04-06	5	Medium	4	3	Healthy	Yes	None	No	0	Naproxen	1

- **Visualize users with pain medium-high**

```sql
SELECT
    U.name,
    AVG(D.pain_intensity) AS avg_pain
FROM USER U
JOIN CYCLE C ON U.user_id = C.user_id
JOIN CYCLE_DAY D ON C.cycle_id = D.cycle_id
GROUP BY U.name
HAVING AVG(D.pain_intensity) >= 6;
```

	name	avg_pain
1	Ana Lopez	8.5
2	Daniela Romoleroux	8.0
3	Marianna Benavides	6.0